DOCUMENT 00 · ELECTRIC HOST ENGINEERING BASELINE

MCRS FULL ELECTRIC

Modular Captured Ride System · CMFSB

Captured Mechanical Floating Safety Body

Two battery-electric host variants · four towers on both · COMFORT / SPORT captured upper body · 275 kWh (V-L) / 300 kWh (V-H) usable energy for ≥800 km design-range capability

Field	Controlled value
Document ID	MCRS-FE-DOC-00-R7.09
Issue date	2026-10-12
Founder / Legal Owner	Amirbahador Atai (individual rights holder; assignment to a legal entity is a separate controlled act)
Programme / IP name	MCRS FULL ELECTRIC · architecture CMFSB · licensable module MCRS-MOD-001
Classification	T1 Confidential / NDA controlled
Document class	Electric-host baseline; module values governed exclusively by Master MCRS-DOC-00-R8.09
Supersedes	MCRS-FE-DOC-00-R7.08. Review R7.09 (RX-01 to RX-06) applied: G1 / G2 exit criteria include REQ-E02, REQ-E03 and REQ-M32 to REQ-M36; release readiness and contract use (Appendix G); owner decisions D-01 to D-08 unchanged.
Status	Proposed baseline, pending owner release and independent technical review. Evidence level L0, TRL 2–3.
Independent Technical Reviewer	To be appointed no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit.
Approval authority	TBD after independent review

This document is a design baseline and diligence instrument. It does not certify range, crashworthiness, homologation, ASIL, production readiness, patentability or freedom to operate. OPEN items remain OPEN until evidence closes them. Module values are controlled exclusively by Master MCRS-DOC-00-R8.09; this document controls host values only.

Contents

1. Executive summary and diligence position . . . 3
2. Basis of values and assumption register . . . 4
3. Product definition and architecture . . . 6
4. Safety invariants and safety layers . . . 8
5. Variant definition and packaging gate . . . 9
6. Electric powertrain and HV architecture . . . 11
7. Energy and range model (V-L 275 kWh / V-H 300 kWh) . . . 13
8. Controlled mass budget . . . 14
9. MCRS module: towers, mass and hydraulics . . . 15
10. MCRS electrical and auxiliary energy . . . 16
11. Control system, modes and state machine . . . 16
12. Power-loss behaviour and lateral-g policy . . . 17
13. Structural load cases and progressive stops . . . 18
14. Controlled requirements register (selected) . . . 19
15. Safety analysis and failure screening . . . 19
16. Verification plan and evidence ladder . . . 19
17. Gates, work packages, risks and open items . . . 21
18. Controlled document tree . . . 22
19. IP, disclosure and commercial paths . . . 22
20. Claims discipline, glossary and release . . . 23
Appendix A — Correction log . . . 24
Appendix B — Status tag definitions (normative) . . . 26
Appendix C — Work package summaries (electric host) . . . 27
Appendix D — Parameter ID cross-reference . . . 28
Appendix E — Release and distribution control . . . 28
Appendix F — Design and engineering review (R7.09) . . . 29
Appendix G — Release readiness and contract use (electric host) . . . 33

1. Executive summary and diligence position

MCRS FULL ELECTRIC is a clean-sheet battery-electric vehicle architecture built around one differentiating subsystem: a mechanically captured, isolated upper safety body. The upper CFRP cell rides on four capture towers above a lower EV chassis that carries the high-voltage battery, e-axles, brakes, steering and thermal systems. Each tower provides large-stroke hydraulic leveling, a force-bounded Lorentz micro-stage, progressive mechanical stops and a SPORT positive lock. The product promise: always captured; soft when comfort is wanted; locked when control is needed; settles on passive support when power is gone.

Controlled item	V-L FULL ELECTRIC	V-H FULL ELECTRIC	Status
Architecture	BEV 4×4, four towers (Class S)	BEV 6×6, four towers (Class L)	BASELINE
Mission class	Urban / grand touring / specialty	Heavy / off-road / defence-utility / specialty	BASELINE
Seats / occupant policy	4 / 400 kg	6 / 600 kg	BASELINE
Overall height (SPORT)	1,950 mm	2,130 mm	FIXED-FOR-ENGAGEMENT
Battery usable / gross (≈90 %)	275 kWh / ≈306 kWh	300 kWh / ≈333 kWh	TARGET / DERIVED
Design-range capability	≥800 km (≈1,060 km at 100 km/h)	≥800 km utility (≤≈72 km/h)	TARGET
Curb mass target	3,544 kg	5,695 kg	TARGET
Design GVW / ceiling	4,044 kg / 4,200 kg	7,000 kg (ceiling; payload residual 705 kg)	TARGET / LOCKED
Peak / continuous traction	≈350–380 / ≈200–220 kW	≈480–550 / ≈280–340 kW	TARGET
Evidence level	L0 architecture	L0 architecture	BASELINE

What this baseline establishes

- A single, arithmetically verified mass, energy and load basis for both variants with 275 kWh (V-L) and 300 kWh (V-H) usable as the primary energy figures.
- Safety invariants independent of the Lorentz loop and of software: capture, passive support, progressive stops, positive locks.
- A re-baselined packaging stack (0 mm V-L, +10 mm V-H) that is explicitly conditional on unvalidated levers.
- Transparent road-load screening showing under which cycles ≥800 km holds.
- A governed route to evidence: gates G0–G5, L0–L5 ladder, document tree, claims-discipline table.
- Explicit subordination to Master MCRS-DOC-00-R8.09 for all module values.

1.1 Readiness assessment

Dimension	Current position	Principal exposure	Closure vehicle
Architecture	Defined, consistent, L0	No hardware or simulation evidence	SIM-FE-001 (G1), tower bench (G2)
Energy / range	275 kWh (V-L) / 300 kWh (V-H) usable; screening model	Pack specific energy 250 Wh/kg is aggressive; V-H 800 km only at low speed; CdA scaled with height	Supplier pack data, ENE-FE-001, CFD
Mass	P50 with reserve	200 Wh/kg adds ≈306 kg (V-L) / ≈333 kg (V-H); V-L margin to 4,200 kg is +156 kg (+43 kg with Class S at 393 kg) and turns to −150 kg at 200 Wh/kg	Supplier BOM, G1 Class S study
Packaging	Stack closes at 0 / +10 mm	Depends on 170 mm of unvalidated levers	G1 3-D datum stack
Safety	Invariants + screening FMEA	No HARA / FTA yet	SAF-FE-001 (G2)
Actuator	Km = 63 N/√W assumption	Thermal and force-gap unproven	ACT-FE-001 (G2)
IP	Ownership chain OPEN	No filing receipt	Counsel, provisional
Commercial	Path A Option defined	No validated interface set	ICD-FE-001 draft (G1)
Regulatory	V-L design GVW 4,044 kg crosses 3,500 kg threshold	Market restrictions vary	REG-FE-001 (G1)

2. Basis of values and assumption register

This baseline consolidates prior electric baselines with 275 kWh (V-L) and 300 kWh (V-H) usable energy (≥800 km design-range capability). Every value was screened for arithmetic and physical consistency. Module values are mirrored from Master R8.09 by citation only.

2.1 Revision history

Revision	Scope
R7.00 – R7.02	Prior consolidated electric baselines (216 / 300 kWh split)
R7.03	Two content passes (Corrected, Fixed) under one ID: 300 kWh both variants; mass re-summed (V-L 3,652; V-H 5,867 kg); P90 range; GVW definition; status tags
R7.04	New ID. Owner decisions D-01 ceiling 4,200 kg, D-02 four V-H towers (module 565 → 393 kg, curb 5,695 kg), D-03 heights 1,950 / 2,130 mm; ranges recomputed with CdA scaled to height; DC fast charge ≥250 kW both; undefined references removed; Master reference R8.04
R7.05	New ID. Owner decision D-05: body zoning and load boundary (§3.4, mirrored from Master R8.05): only the passenger cell is suspended; battery in the lower chassis adds no load to the cell or module; payload-location sensitivity; seam envelope; Master reference R8.05
R7.06	New ID. Owner decisions D-06 (V-L usable battery 275 kWh, gross ≈306 kWh, pack 1,224 kg; V-H stays 300 kWh) and D-07 (only the safety cockpit is suspended; hood, trunk and payload on the lower body). V-L curb 3,544 kg, GVW 4,044 kg, margin +156 kg; ranges, charging, loads and modes recomputed; assumption A-15 (lower bodywork); Master reference R8.06
R7.07	Review of R7.06: consistency items EC-03, EC-05, EC-07, EC-09, EC-10; engineering findings EN-01 to EN-07; REQ-E02 and REQ-E03; Master reference R8.07
R7.08	New ID. Stop-stroke efficiency and design speed, horizontal capture bound, V-L pack sizing rule, neutral ceiling wording, height / lever / seam sensitivities (Appendix F); mass-table note corrected; supersession chain restated; page count above the previous issue; Master reference R8.08
R7.09 (this issue)	New ID. Review RX-01 to RX-06: G1 / G2 exit criteria include the new requirements; assumption register and release-readiness statement (Appendix G); Master reference R8.09

2.2 Assumption register (each item requires owner approval)

ID	Assumption	Value	Effect if wrong	Closure
A-01	Pack-level gross specific energy	250 Wh/kg (aggressive)	Mass +≈306 kg (V-L) / +≈333 kg (V-H) at 200 Wh/kg	Supplier pack data G1/G2
A-02	Usable fraction of gross	≈90 %	Range scales linearly	BMS / supplier
A-03	CdA (scaled with height), Crr, drivetrain η, aux	0.99 / 1.6 m²; 0.009 / 0.012; 0.88 / 0.87; 1 / 2 kW	Range ±10–20 %	ENE-FE-001, CFD
A-04	Reference test mass	Curb + 200 kg	Consumption shifts	SIM-FE-001
A-05	Programme mass reserve	≈3.9–4.0 % of curb	Insufficient for supplier growth	Mass closure G2
A-06	Primary usable energy	275 kWh V-L / 300 kWh V-H	Range and mass scale	Owner decision
A-07	Occupant policy	100 kg each	Upper mass, GVW	Programme policy
A-08	Class S corner mass target	70 kg (module 280 kg)	+96 kg (corner) / +113 kg (with distribution) if 94 kg	Class S study G1
A-09	Packaging levers adopted as allocation	170 mm total	Residual becomes negative	G1 3-D stack
A-10	V-H utility cycle	≈55–60 km/h avg	800 km not met at highway speed	Cycle definition G1
A-11	K-eff basis	40 kN/m per tower	Natural frequency	ACT / STR-FE-001
A-12	Charging average power	63 % / 60 % of peak	Charging time	Supplier charge curve
A-13	RPN values	Inherited screening	Priorities may change	SAF-FE-001 rescoring
A-14	Height increase mass-neutral	+250 / +130 mm	Mass and CG shift	G1 SIM
A-15	Lower bodywork mass (hood, wings, trunk, rear body)	Inside the chassis group; not itemised	Mass and CG shift	G1 mass freeze

2.3 Reference host comparison (with Hybrid R5.40)

Item	V-L electric	V-L hybrid	V-H electric	V-H hybrid
Powertrain	BEV dual e-axle	ICE V8 4.0 L TT + e	BEV 6×6 three e-axles	ICE V12 6.0 L TT + e
On-board energy	275 kWh usable	80 L + 20 kWh	300 kWh usable	150 L + 40 kWh
Range target	≥800 km	1,000 km ≤70 km/h	≥800 km utility	700–800 km
Towers	4 (Class S)	4 (Class S)	4 (Class L)	4 (Class L)
Curb target	3,544 kg	2,778 kg	5,695 kg	4,983 kg
Design GVW	4,044 kg	3,278 kg	7,000 kg	5,983 kg
Margin to ceiling (4,200 / 7,000)	156 kg	922 kg	0 kg	1,017 kg
Stack residual	0 mm	0 mm	+10 mm	+10 mm
Evidence	L0	L0	L0	L0

3. Product definition and architecture

The host is defined around the CMFSB load and isolation architecture; it is not a powertrain retrofit. The MCRS tower set is the sole mechanical connection between the two body pieces in normal operation. Active isolation works inside a bounded mechanical safety envelope; comfort control, handling stiffness and crash containment are coordinated but independently bounded.

3.1 Layer stack (top to bottom)

Layer	Function	Safety role
Upper safety cell	Occupant and mission body (CFRP, roll structure, seats, closures)	Captured at all times
Micro stage (Lorentz)	High-bandwidth isolation within ±7.5 mm	Force-bounded; never a crash or horizontal path
Macro stage (hydraulic)	Leveling, ±50 mm body-relative travel, lock seats	Maintains geometry and ride envelope
Progressive stops	Energy absorption on abnormal vertical events	Primary abnormal vertical energy path
SPORT positive lock	Pin, claw or hydraulic clamp at the seat band	Handling stiffness and load sharing; not a replacement for stops
Capture frame and links	Longitudinal, lateral and yaw restraint	Primary horizontal path
Lower chassis hardpoints	Load transfer into the host platform	Structural continuity

Figure 1 (conceptual). Two-piece body: the upper cell is captured on MCRS towers above the lower host chassis. V-L: four towers (Class S). V-H: four towers on both hosts (owner decision D-02); six towers remain an option of MCRS-MOD-001. Vertical support is passive macro support; horizontal loads go through the capture frame and links; abnormal vertical energy goes through progressive stops.

3.2 Independent load and force paths

Load / function	Primary path	Secondary / containment
Static vertical support	Passive hydro-pneumatic macro support	Progressive stops / chassis structure
COMFORT dynamic force	Lorentz micro actuator (force-bounded)	Macro stage + passive support
Large displacement, leveling	Hydraulic macro stage	Stops
SPORT stiffness	Positive locks + seat faces	Capture frame
Longitudinal / lateral / yaw	Capture frame, links, horizontal stops	SPORT lock faces
Crash / abnormal vertical	Progressive stops + structural hardpoints	Capture frame / links
Power loss	Passive hydraulic / compliant support (S4)	Mechanical capture
Battery crash load	Lower chassis + protected HV enclosure	Independent containment structures

The Lorentz stage is an actuator option. If another actuator family replaces it, the safety invariants and interface requirements still apply under configuration control. Hold is an optional, rate-limited secondary restraint (orifice and/or friction at the macro stage); it is distinct from the SPORT lock and the stops and is not a primary crash path.

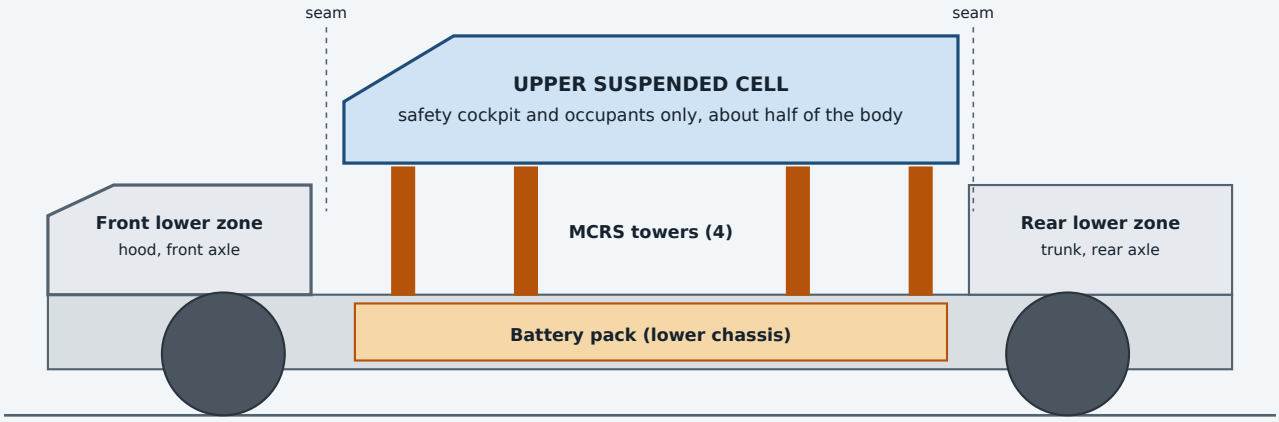
3.3 Qualitative comparison

Approach	Comfort	Handling stiffness	Positive lock	Independent capture	Power loss
Soft air / hydro-pneumatic	Good	Reduced in rapid manoeuvres	No	Limited to suspension travel and bump stops	Settles on bump stops
Adaptive dampers	Improved trade-off	Improved	No	No	Fixed or fail-soft damping
CMFSB / MCRS (target)	Targeted (unproven at L0)	Targeted via SPORT lock	Yes (SPORT)	Yes: stops, frame, links	Settles on passive support, captured

Qualitative only; no measured superiority is claimed at evidence level L0.

3.4 Body zoning and load boundary (owner decisions D-05, D-07)

Only the safety cockpit is suspended: the passenger cell with the occupants' space, about half of the overall body (planning ≈2,500 mm of the 5,000 mm V-L body and ≈2,750 mm of the 5,500 mm V-H body). The hood, front wheel housings, trunk, rear body and wheel housings are attached to the lower body, and the lower chassis forms a cradle around the cell. The battery, e-drive or ICE, fuel tank, axles, brakes, steering and the trunk payload are all on the lower chassis.



Battery, powertrain, fuel, hood and trunk payload weigh on the lower chassis only. The module carries only the cockpit.

Figure 1b (conceptual, not to scale). Side view of the zoning: the upper cell rides on the MCRS towers; battery, powertrain and axles belong to the lower chassis.

Zone	Contents	Carried by	Loads the module?
1 Upper suspended cell (safety cockpit)	CFRP safety cell, roll structure, seats, closures, occupants (no payload, no battery)	MCRS towers	Yes: this is the upper sprung mass
2 Front lower zone	Hood, front wheel housings, front axle, front crash structure	Lower chassis	No
3 Rear lower zone	Trunk, rear body and wheel housings, rear axle(s), payload	Lower chassis	No
4 Lower chassis	Battery pack (electric V-L 275 kWh / V-H 300 kWh; hybrid 20 / 40 kWh), e-drive / ICE, fuel tank, brakes, steering, thermal system	Wheels and suspension	No: never loads the module

How the battery relates to the upper level. It adds no load to the module or the cell. It matters only through (a) total vehicle weight against the GVW ceiling and (b) its thickness inside the lower chassis, which sets the chassis height in the packaging stack (pack volume ≈1,020–1,665 litres). The tower depth shared with the chassis and battery floor (60 mm lever) is the only place where battery packaging and module packaging overlap.

Seams. The joints between the moving cell and the fixed front and rear bodywork shall accommodate ±57.5 mm of normal relative motion (50 mm macro + 7.5 mm micro) plus the stop stroke (50 mm placeholder) without contact, using flexible seals or bellows; values are frozen in ICD-001. Seam water, dust, noise and pinch protection is not yet designed (OPEN).

Payload location (decision D-07). Payload is carried in the trunk on the lower body and does not load the module; the upper sprung mass is the cockpit structure plus occupants (and the optional protection package where fitted). Had payload been carried in the cabin, the upper mass would rise by 100 kg (V-L) and 705 kg (V-H electric), and the V-H electric 20 g corner load from ≈101 to ≈135 kN. Trunk payload loads the lower chassis and rear axle instead (STR-001, G1).

4. Safety invariants and safety layers

§4 is the controlling statement of the safety invariants. Appendix I and both host baselines mirror it; the mirrors are labelled and §4 prevails on any difference.

Invariant	Rule
Mechanical capture	No operating state permits loss of geometric capture; no free levitation.
Static support	Vehicle weight on passive or fail-safe compliant paths, never the sole magnetic path.
Power loss (S4)	Passive support; powered leveling-supply and hold valves fail-closed; the hydro-pneumatic spring path (cylinder to accumulator) stays connected; software aliveness not required.
Crash path	Progressive stops + structure, independent of closed-loop magnetic force.
Mode transition	SPORT↔COMFORT never enters unconstrained free-float.
SPORT lock fault	Lock failure shall not defeat capture or progressive stops; SPORT is never declared with an incomplete lock set.
Horizontal restraint	Longitudinal / lateral / yaw loads resolved by dedicated paths and/or SPORT lock.
Override limit	Driver override cannot defeat the lateral-g interlock or a critical fault state.
Manual rescue	Two-step rescue available under power loss and hydraulic failure.
HV isolation	Traction HV galvanically isolated from the module control domain; a single HV fault shall not defeat capture or S4.
Thermal / fire	Battery runaway containment is a lower-chassis responsibility and shall not propagate into upper-body capture integrity.
Single-path rule	No single active-control path is credited as the primary crash path.
Lock state on power loss	Locks remain engaged only if already engaged at the instant of loss (SPORT). From COMFORT the locks stay released. Logic frozen in ICD-001 / LCK-001 (REQ-M17 / M20).

4.1 Layered safety response

Safety layer	Failure addressed	Design response
Mechanical capture	Link / carriage failure	Capture frame with redundant restraint paths
Passive support	Power loss	Hydro-pneumatic / compliant support, fail-safe valves
Progressive stops	High-energy vertical event	Defined force–displacement curve (STR-001)
Horizontal restraint	Braking, cornering, crash	Dedicated links, stops and lock faces
Positive SPORT locks	Mode stiffness, lock integrity	Per-lock confirmation; incomplete set inhibits SPORT
HV safety	Electrical crash, isolation fault	HVIL, isolation monitor, service disconnect, crash inhibit
Thermal safety	Battery / inverter overheating	Redundant sensing, derating, propagation-resistant pack, vent routing
Control safety	Bad command, software fault	Bounds, diagnostics, degraded state, independent monitor

No ASIL assignment is claimed. High-severity (S ≥ 9) items require explicit pre-G2 review independent of RPN score.

Mirrored from Master §4 and §4.1; the Master governs on any difference.

5. Variant definition and packaging gate

5.1 Geometry and mission

Parameter	V-L	V-H
Length / width (mirrors excl.)	5,000 / 2,000 mm	5,500 / 2,200 mm
Overall height (SPORT)	1,950 mm (re-baselined)	2,130 mm (re-baselined)
Wheelbase	3,000 mm	3,300 mm (front axle to mid of rear tandem)
SPORT ground clearance	200 mm	300 mm
Drive	4×4 electric	6×6 electric (front + dual rear tandem)
Seats	4	6
Body	CFRP safety cell + structural roll frame (FIA-referenced philosophy; no homologation claim)	CFRP safety cell + structural roll frame
MCRS class / towers	Class S / 4	Class L / 4 (six-tower stays a module option)
GVW engagement ceiling	4,200 kg	7,000 kg
Optional light protection package	—	≈200 kg; identical treatment to hybrid

Dimensions are programme targets: FIXED-FOR-ENGAGEMENT / PACKAGING GATE OPEN. V-H is not a scaled V-L. Status tags are defined in Appendix B.

5.2 Vertical packaging stack (P0 open item)

Element (mm)	V-L	V-H	Status
Overall height (SPORT)	1,950	2,130	FIXED-FOR-ENGAGEMENT
Ground clearance (SPORT)	200	300	FIXED-FOR-ENGAGEMENT
Available for structure (height − GC)	1,750	1,830	DERIVED
Chassis / axle structure	280	320	STUDY
Tower structure + mounts (60 mm shared with chassis)	60	80	STUDY
Macro travel (100) + free play (10)	110	110	STUDY
Stop stroke + micro-stage clearance (placeholder)	50	50	ASSUMPTION
Floor-to-H-point	220	220	STUDY
Cabin, H-point to roof	960	960	TARGET
Roof / exterior shell	70	80	STUDY
Allocated sum	1,750	1,820	DERIVED
Residual margin	0	+10	OPEN — P0 (conditional)

Closure lever adopted (mm recovered)	V-L	V-H	Basis
Share tower depth with chassis / battery floor (120 → 60 / 140 → 80)	60	60	Integration study
Reduce travel + free play (140 → 110; free play 40 → 10)	30	30	Stop-stroke trade
Cabin H-point to roof 1,000 → 960	40	40	Ergonomic package
Floor-to-H-point 250 → 220	30	30	Seat / floor design
Shell thickness (80 → 70 / 90 → 80)	10	10	CFRP roof design
Total recovered	170	170	

Heights were raised from 1,700 / 2,000 mm to 1,950 / 2,130 mm (owner decision D-03). Residual 0 mm (V-L) and +10 mm (V-H) holds only if every lever is delivered; none is validated, V-L has zero margin, and the 50 mm stop-stroke allowance is a placeholder until STR-001. Status stays OPEN — P0 until the G1 3-D datum stack confirms it.

6. Electric powertrain and HV architecture

Item	V-L	V-H	Status
Nominal HV bus	800 V class	800 V class	TARGET
Drive topology	Dual e-axle AWD	Front + dual rear e-axles (three)	TARGET
Front e-motor(s)	≈175–190 kW	≈150–180 kW	TARGET
Rear e-motor(s)	≈175–190 kW	≈330–370 kW (dual)	TARGET
Peak traction power	≈350–380 kW	≈480–550 kW	TARGET
Continuous traction	≈200–220 kW	≈280–340 kW	TARGET
On-board charger (AC)	22 kW base / 44 kW option	22 kW base / 44 kW option	TARGET
DC fast charge	≥250 kW class	≥250 kW class	TARGET (same class on both variants)
Battery usable / gross	275 kWh / ≈306 kWh	300 kWh / ≈333 kWh	TARGET / DERIVED

Axle splits are consistent with the published peaks (V-L each axle ≈175–190 kW → 350–380 kW; V-H rear dual ≈330–370 kW + front → 480–550 kW).

6.1 HV safety functions

HV function	Baseline requirement
Battery enclosure	Protected, structural, serviceable, crash-contained; does not rely on upper body for primary containment
HVIL	Continuous loop over pack, service disconnects, power electronics and charge interface
Isolation monitoring	Continuous during energized operation; fault response in ICD-FE-001
Precharge	Controlled before inverter DC-link energisation
Crash isolation	Independent crash-signal path plus BMS / contactor logic
Service disconnect	Physically accessible and lockable
Thermal propagation	Cell-level detection, propagation-resistant pack, vent routing
HV-to-MCRS separation	MCRS safe-state support remains available if traction HV is isolated
LV backup	Dedicated auxiliary energy for controlled safe-state functions

6.2 Detailed HV architecture notes

The 800 V class bus is selected for efficiency and charge-power capability. A single HV fault shall not defeat MCRS capture or S4. Fast-charge thermal limits are not yet quantified (THERM-FE-001); published DC charge rates remain TARGET and may be derated by thermal duty. Both variants carry the same ≥250 kW class; at the A-12 average of 60 % a 10→80 % charge takes about 1.3 h (V-L, 275 kWh) and 1.4 h (V-H, 300 kWh). The module 48 V domain remains galvanically isolated from the host HV bus.

6.3 e-drive topology notes

V-L dual e-axle AWD provides balanced traction and regeneration. V-H three-axle (front + dual rear tandem) matches the 6×6 mission and allows axle-selective torque. Inverter and e-motor suppliers are not frozen (ICD-FE-001). Regeneration coordinates with the brake controller; MCRS mode does not alter the primary brake path.

7. Energy and range model (V-L 275 kWh / V-H 300 kWh)

Range = usable energy ÷ average consumption. Primary usable energy is 275 kWh (V-L) and 300 kWh (V-H). All screening is first-order; no regeneration, grade or temperature credit. At a pack-level 200–300 Wh/L the ≈306 / ≈333 kWh gross packs occupy ≈1,020–1,665 litres; packaging them in the lower chassis is part of the G1 datum stack.

7.1 Energy budget

Item	V-L	V-H
Gross battery energy	≈306 kWh	≈333 kWh
Reserved / inaccessible (≈10 %)	≈31 kWh	≈33 kWh
Net usable energy	275 kWh	300 kWh
Required average for 800 km	344 Wh/km	375 Wh/km
Required average for 900 km	306 Wh/km	333 Wh/km

7.2 Screening assumptions

Assumption	V-L	V-H	Note
Reference mass (curb + 200 kg)	3,744 kg	5,895 kg	Revised for four-tower V-H
CdA	0.99 m²	1.6 m²	Scaled with height, ASSUMPTION
Crr	0.009	0.012	Road / off-road
Battery-to-wheel η	0.88	0.87	Drive + inverter
Auxiliary load	1 kW	2 kW	Constant

Method: $F = ½·\rho·CdA·v^2 + Crr·m·g$; wheel energy = F ÷ 3.6 (Wh/km); battery energy = wheel energy ÷ η + P_aux ÷ v; ρ = 1.2 kg/m³.

7.3 Range screening results (V-L 275 kWh / V-H 300 kWh usable)

Case	Approx. total Wh/km	Range on usable energy
V-L steady 80 km/h	≈209	≈1,310 km
V-L steady 100 km/h	≈259	≈1,060 km
V-L steady 120 km/h	≈321	≈860 km
V-H utility 55–60 km/h	≈329–340	≈880–910 km
V-H 90 km/h	≈435	≈690 km
V-H 120 km/h	≈579	≈520 km

Design-range statement. V-L keeps ≥800 km at steady speeds up to ≈126 km/h. V-H supports ≥800 km only on a low-speed utility cycle (up to ≈72 km/h steady); at highway speeds range falls to ≈520–690 km. Cycle definition is frozen in SIM-FE-001 / ENE-FE-001. The earlier R7.03 bands for V-H utility (≈880–1,000 km) and V-L 120 km/h (≈910–1,000 km) were optimistic and are replaced.

7.4 Charging times

AC: 22 kW base → 0→100 % ≈12.5 h (V-L 275 kWh) / ≈13.6 h (V-H 300 kWh); 44 kW option → ≈6.3 h / ≈6.8 h. DC fast-charge times depend on charge curve and thermal limits (THERM-FE-001).

7.5 Pack mass sensitivity

Specific energy (gross)	V-L pack (≈306 kWh gross)	V-H pack (≈333 kWh gross)	Effect
250 Wh/kg (baseline ASSUMPTION)	≈1,224 kg	≈1,332 kg	Baseline
200 Wh/kg (conservative)	≈1,530 kg (+306)	≈1,665 kg (+333)	V-L margin −150 kg; reserve consumed; re-baseline
180 Wh/kg	≈1,700 kg (+476)	≈1,850 kg (+518)	Significant mass growth

250 Wh/kg pack-level is an aggressive ASSUMPTION (current production packs are typically 140–180 Wh/kg; high-end claims approach 200–250). Mass, drag and battery mass are coupled.

8. Controlled mass budget

Curb = arithmetic sum of groups + explicit reserve. V-H design GVW is set equal to the 7,000 kg ceiling; payload is the balancing residual. For V-L, design GVW = curb + occupants + payload. Payload policy is decided at G1. All figures TARGET / STUDY at L0.

Group	V-L (kg)	V-H (kg)	Note
Upper body (CFRP cell + roll + seats + closures)	780	1,250	Cockpit structure; payload, battery and hood / trunk excluded
Battery pack (V-L 275 kWh ≈306 kWh gross; V-H 300 kWh ≈333 kWh gross; @ 250 Wh/kg)	1,224	1,332	Primary energy
Chassis, axles, wheels, brakes, steering	820	1,800	
e-drive, inverters, HV distribution	180	280	
MCRS module (four towers; P50)	280	393	Class S / Class L
HV/LV, cooling, thermal	120	220	
Optional protection package	—	200	Identical treatment to hybrid
Reserve	140 (≈4.0 %)	220 (≈3.9 %)	Explicit
Curb total (TARGET)	3,544	5,695	
Occupants (policy)	400	600	100 kg each
Payload (V-L policy / V-H residual)	100	705	V-H residual to 7,000 kg ceiling; carried in the trunk (no module load)
Design GVW	4,044	7,000	V-H = ceiling
Ceiling	4,200	7,000	LOCKED
Margin	156	0	DERIVED

Check: V-L 780 + 1,224 + 820 + 180 + 280 + 120 + 140 = 3,544; 3,544 + 400 + 100 = 4,044. V-H 1,250 + 1,332 + 1,800 + 280 + 393 + 220 + 200 + 220 = 5,695; 5,695 + 600 + 705 = 7,000. The 275 kWh V-L pack saves 108 kg against 300 kWh: design GVW 4,044 kg is within the 4,200 kg ceiling by 156 kg (D-01, D-06) and above the 3,500 kg threshold (REG-FE-001). With the Class S module at 393 kg the margin is still +43 kg; at 200 Wh/kg pack specific energy the V-L pack grows by ≈306 kg and the design is 150 kg over the ceiling. The four-tower decision (D-02) cuts the V-H module from 565 to 393 kg and raises the V-H payload residual from 533 to 705 kg; payload rides in the trunk.

P90 MCRS mass (V-H, four towers). RSS ≈427 kg (pump independent), ≈439 kg (pump correlated), arithmetic 565 kg, against P50 393 kg (+34 / +46 / +172 kg). The 220 kg reserve covers all three cases and leaves 48 kg in the arithmetic case; it is shared with every other group. Risk R-BATT is retired.

Upper sprung mass (cockpit structure + occupants; payload rides in the trunk and does not load the module): V-L ≈1,180 kg; V-H ≈2,050 kg (with protection) or ≈1,850 kg (without).

9. MCRS module: towers, mass and hydraulics

9.1 Corner assembly mass (Class L planning)

Element	P50 kg	P90 kg	Deviation (P90 − P50)	Scope
Local structure (per tower)	18	25	7	Core
Lorentz + local carriage	22	30	8	Core
Sensors + local electronics	8	12	4	Core
Core tower subtotal	48	67	RSS 11.4	Target ≤55 kg P50 met by estimate; limit until G2 = 67 kg
Cylinder + accumulator share	15	22	7	Corner
Valves / hoses / seals	5	8	3	Corner
Progressive stops	8	12	4	Corner
SPORT lock share	6	10	4	Corner
Harness / dampers / bellows	4	6	2	Corner
Pump + reservoir share (¼)	8	12	4 (16 for the shared pump)	Shared
Full corner assembly (Class L)	94	137		STUDY; ≈95 kg in host WBS

Core-tower RSS is $48 + \sqrt{(7^2 + 8^2 + 4^2)} = 59.4$ kg. The 67 kg arithmetic value stays the limit until G2 supplier correlation data; a 60 kg limit is accepted only with correlation evidence.

9.2 Module mass decomposition and P90 method

Authoritative decomposition for host mass tables: dedicated corner 86 / 125 kg (P50 / P90), shared pump 32 / 48 kg, distribution 17 kg (no P90 deviation assumed). Four-tower Class L P50 = 4 × 86 + 32 + 17 = 393 kg; six-tower P50 = 6 × 86 + 32 + 17 = 565 kg (one shared pump in both).

P90 method. Per-element deviations of the dedicated corner (7, 8, 4, 7, 3, 4, 4, 2 kg) combine to 14.9 kg RSS per corner. Corners are assumed independent: $\sqrt{n} \times 14.9$. The shared pump deviation (16 kg) is one further independent term in the same quadrature sum, so the deviation is $\sqrt{(n \times 223 + 16^2)}$. It is not added linearly.

Case	4-tower Class L	6-tower Class L	Meaning
P50	393 kg	565 kg	Budget value
P90, RSS (pump independent)	427 kg (+34)	605 kg (+40)	Method above; independence assumed
P90, RSS (pump fully correlated, linear)	439 kg (+46)	618 kg (+53)	Sensitivity; sensible upper RSS bound
P90, arithmetic (fully correlated)	565 kg (+172)	815 kg (+250)	Worst case: 4 × 125 + 48 + 17; 6 × 125 + 48 + 17

Reserve coverage: hybrid V-H reserve 150 kg covers both RSS cases but not the arithmetic case (+172 kg). Electric V-H (four towers, module 393 kg) reserve 220 kg covers all three cases, leaving 48 kg in the arithmetic case; the reserve is a programme reserve shared with every other group. Corner independence is itself an assumption until G2 supplier correlation data. Six towers (a module option, not baselined on either host) add ≈172 kg over four towers (565 − 393); the 605 / 618 / 815 kg figures apply to that option only.

9.3 Per-class corner mass and static support floor

Class	Mapped use	Corner P50	Module P50	Static basis per corner	Floor (1.25 × Fmax)
S	Light host (four seats)	≈70 kg (target)	≈280 kg (4 × 70, corner-only; distribution allocation OPEN)	Occupied share 310–350 kg	≥4.5 kN
M	Intermediate	≈85 kg	≈340 kg	Share 450–550 kg	≥7.0 kN
L	Heavy host	≈95 kg (94 incl. pump share)	393 kg (4 towers) / 565 kg (6 towers), incl. 17 kg distribution	Reference max 787.5 kg	≥10.0 kN

Class S exposure. The 70 kg corner has no design evidence. If a Class S corner equals the Class L architecture (94 kg), the four-tower module grows by 96 kg (corner-only) to 113 kg (with the 17 kg distribution). The hybrid V-L reserve (113 kg after D-08) and the electric V-L reserve (140 kg) both cover it; with the 275 kWh pack the electric V-L design GVW stays 43 kg inside the 4,200 kg ceiling. With payload in the trunk the V-L occupied share per corner is ≈295 kg ((780 + 400) ÷ 4), below the 310–350 kg basis; the 4.5 kN floor is kept as the conservative basis. OPEN: G1 Class S study (REQ-M30).

10. MCRS electrical and auxiliary energy

Module supply is 48 V class, galvanically isolated from host HV. Peak coil power per tower is $(F \div Km)^2 = (2{,}500\ N \div 63\ N/\sqrt{W})^2 \approx 1.575$ kW, so the micro-stage array peaks at ≈6.3 kW (four towers) and ≈9.45 kW (six towers). These are peak, not RMS, values and exceed the baseline DC-DC ratings of 3 kW and 4 kW. A local energy buffer (48 V battery or supercapacitor) is therefore mandatory for both configurations. The buffer serves comfort performance only; the S4 safety case does not rely on it. Safe-state energy reserve is verified under ENE-E-001 (electric) and ENE-H-001 (hybrid); power-loss detection and valve-closure timing are frozen in ICD-001.

Domain	Voltage	Isolation	Function
Module control / Lorentz	48 V class	Galvanic from host HV	Micro actuators, sensors, BCE
Hydraulic valves / hold	48 V / 12 V	Fail-closed on loss	Leveling, hold, lock actuation
Host traction HV	High voltage	Isolated from module	e-drive, HV battery (host responsibility)
Local energy buffer	48 V	Module domain	Peak shaving for Lorentz; mandatory in both configurations

11. Control system, modes and state machine

11.1 Controller partition

Domain	Responsibility
Vehicle domain controller	Driver intent, drive state, dynamics, high-level requests; cannot bypass MCRS interlocks
MCRS Body Control ECU (BCE)	Mode arbitration, corner coordination, command bounds, lock sequencing, leveling, diagnostics
Independent safety monitor	Force / stroke / mode bounds, critical interlocks, watchdog and plausibility, where adopted
Corner controllers	Local sensing and actuator current / valve commands, bounded by BCE
BMS	Battery state, HV safety, contactor control, thermal limits
Brake controller	Primary service braking and regenerative coordination

Comfort control may be sophisticated; safety bounds remain simple, deterministic and independently monitorable. The monitor may be called independent only after power, clock, processor, software, communication and sensor dependencies are analysed and tested. OTA or calibration changes shall not bypass configuration control.

11.2 State machine

State	Name	Rule
S0	Safe / captured / power-off	No powered release; passive capture retained (equivalent to S4)
S1	Precharge / self-test	Capture healthy; auxiliary power path checked
S2	COMFORT	Macro + micro isolation active; capture mandatory
S2S	SPORT	Seat band; all required positive locks confirmed; stages held
S3	Degraded	Reduced authority; force and travel bounded
S4	Power-loss capture	Passive support; fail-safe valves; no software dependency for retention
S5	Impact preparation	High-rate unlock inhibited; stops prepared
S6	Impact / arrest	Progressive stops active; no actuator authority required
S7	Post-event safe	No automatic re-enable; manual clearance required
S8	Maintenance lockout	Physical lockout; stationary only

Any powered state goes to S4 on power loss (engaged locks stay engaged only if already locked; from COMFORT they stay released). S4 exits only to S1 after power restore and self-test. S2↔S2S only under the lateral-g policy with complete lock confirmation. The full transition table, including S5–S8 exit conditions, is required in ICD-001. Detection of power loss ($t_{det,max}$) must be hardware/passive so that S4 without software is not contradicted.

11.3 Mode sequences

Transition	Sequence
COMFORT → SPORT	Request accepted (no inhibit, lateral-g met, no fault) → macro drives to seat band → seat contact at required corners → locks engage → all required locks confirmed (incomplete set aborts to COMFORT) → stages held → S2S
SPORT → COMFORT	Allowed only under lateral-g policy → lock release commanded → all report released → macro envelope re-enabled → micro control re-enabled → S2
Power loss from COMFORT	Immediate S4; active force removed; passive support retained; locks stay released
Power loss from SPORT	S4 / fail-locked; locks remain engaged until controlled recovery
Critical fault	S3 / S4 as applicable; no automatic comfort restoration

12. Power-loss behaviour and lateral-g policy

12.1 REQ-M07 power-loss structure

After loss of commanded active power the system shall: (1) detect the loss within t_det,max (hardware/passive path); (2) enter S4; (3) close the relevant valves within t_valve,max; (4) remain inside the capture envelope; (5) stay below z_max and v_max; (6) prevent uncontrolled free-float; (7) settle in the defined S4 hydraulic state with passive support at or above the class static floor, with residual capacity and environment stated, under electrical loss and the defined hydraulic failure cases. State definition alone is not acceptance.

Limit to freeze	Where	Verified by
t_det,max, t_valve,max	ICD-001 (G2)	Bench + fault injection (L2/L3)
z_max, v_max, settle limit	ICD-001 (G2)	Single-tower power-cut test
Residual stiffness and damping in S4	ICD-001 (G1/G2)	L2 bench
Static support floor after hose-burst case	STR-001 (G2)	Hydraulic failure test
Energy reserve for safe-state functions	ENE-E-001 / ENE-H-001	Power-loss timing test

12.2 Lateral-g policy (REQ-M21)

Element	Rule
Inhibit	SPORT→COMFORT inhibited immediately when |ay| > 0.35 g, for automatic and manual requests
Re-enable	Only after |ay| stays below 0.30 g for the hold time
Driver override	May queue a request; cannot defeat the interlock
Filter and latency	Validated filter and measurement-to-inhibit latency budget, both biased to inhibit
Sensor faults	Validity, plausibility, disagreement thresholds, stale-data handling; a single lateral sensor cannot alone trigger an unsafe transition

Thresholds are temporary TARGET values; SIM-001 and HIL-001 freeze final thresholds, hysteresis, filtering and latency. PARAM-M21-ROLL-GRAD is defined at G2.

13. Structural load cases and progressive stops

Screening loads are based on upper sprung mass (cockpit structure + occupants; payload rides in the trunk), not GVW. Corner loads assume equal share (÷4) and are a magnitude screen only; STR-FE-001 must calculate the real distribution with CG, payload placement and load transfer under 1 g braking / cornering.

Case	V-L upper ≈1,180 kg	V-H upper ≈2,050 kg	Note
1 g static, per corner	≈2.9 kN	≈5.0 kN	Four towers; ≈513 kg inside the 787.5 kg Class L reference
2 g vertical	≈5.8 kN	≈10.1 kN	Screening
20 g vertical	≈58 kN	≈101 kN	Stops path
1 g braking transfer	TBD	TBD	STR-FE-001 — not equal share
Cornering transfer	TBD	TBD	STR-FE-001 — CG dependent
Abnormal single-tower	TBD	TBD	Fail-safe load path

Progressive stops provide the primary abnormal vertical energy path; capture frame and links the primary horizontal path; SPORT locks are not a substitute for stops. Stop stroke is a 50 mm placeholder in the packaging stack, tied to a design vertical impact speed in Appendix F.6 (OPEN, REQ-M33).

13.1 Natural frequency screening

With K-eff 40 kN/m per tower (A-11), four-tower Class S on V-L upper mass ≈1,180 kg yields a first vertical mode of ≈1.9 Hz; four-tower Class L on V-H upper mass ≈2,050 kg yields ≈1.4 Hz. Per-tower stiffness for Class L four-tower is OPEN (ACT / STR-FE-001). Control bandwidth target ≥20 Hz remains.

14. Controlled requirements register (selected)

ID	Statement (summary)	Status
REQ-E01 / V01	Primary usable energy 275 kWh (V-L) and 300 kWh (V-H); ≥800 km design capability under a defined cycle	TARGET
REQ-M11	Module interface and ICD compliance (mirrored from Master)	OPEN — ICD-FE-001
REQ-M29 / V04	V-H tower count: four Class L towers on both hosts (decided 2026-10-12)	BASELINE
REQ-M07	Power-loss structure	OPEN — ICD-FE-001
REQ-M17/M20	Lock state on power loss from COMFORT	OPEN — LCK-001
REQ-M21	Lateral-g interlock	TARGET (temporary)
REQ-M30	Class S corner mass basis incl. distribution allocation	OPEN — G1
REQ-M31	Body zoning: only the safety cockpit is suspended; battery, hood and trunk payload on the lower body; seam envelope in ICD-FE-001	BASELINE / seam OPEN — ICD-FE-001
REG-FE-001	Regulatory exposure incl. V-L design GVW 4,044 kg vs 3,500 kg threshold; ceiling 4,200 kg	OPEN — G1
REQ-E02	Seam vent isolation: battery runaway gas shall not reach the cell through the seams; 107.5 mm overtravel (Master M.10)	OPEN — ICD-FE-001 / THERM-FE-001
REQ-E03	V-L energy sizing: usable energy follows the rule of Appendix F.4 (≤ 275 kWh; design GVW with Class S and the specific-energy sensitivity ≤ 4,200 kg), or the owner re-decides at G1 (REQ-M36)	OPEN — Owner / G1
REQ-M32 to M36	Packaging margin ≥ +20 mm, vertical arrest (η ≥ 0.70, 3.0 m/s baseline), horizontal crash path, seam function, ceiling sizing (Master Appendix M)	OPEN — Master App. M

15. Safety analysis and failure screening

Screening FMEA at L0. No HARA / FTA yet. SAF-FE-001 (G2). High-severity (S ≥ 9) rows require pre-G2 independent review regardless of RPN.

ID	Failure mode	Effect	S	O	D	RPN	Action
F-01	Loss of geometric capture	Upper body free	10	2	4	80	Pre-G2; redundant paths
F-02	Link / carriage fracture	Horizontal path lost	9	2	5	90	Pre-G2; STR-FE-001
F-03	SPORT lock incomplete	False SPORT	8	3	3	72	Per-lock confirmation
F-04	Power-loss valve fails open	Loss of passive support	9	2	4	72	Fail-closed; test
F-05	HV fault to module	Capture defeated	9	2	5	90	Galvanic isolation; HVIL
F-06	Thermal event near cell	Propagation risk	8	3	4	96	THERM-FE-001
F-07	Single lateral sensor fault	Unsafe transition	7	3	4	84	Plausibility + dual path
F-08	Software out of bounds	Force exceedance	7	3	3	63	Independent monitor
F-09	Pack thermal runaway	Lower chassis event	9	2	4	72	Propagation-resistant design
F-10	Buffer depletion under peak	Comfort loss only	4	3	3	36	ENE-FE-001 sizing

Severity ≥9 rows flagged for mandatory pre-G2 independent review regardless of RPN. Full FMEA in SAF-FE-001.

16. Verification plan and evidence ladder

Current position L0 (architecture). SIM-001 (G1) and the tower bench (G2) are the first simulation and hardware vehicles. All state-machine transitions require HIL evidence before credit.

Level	Meaning	Typical artefacts
L0	Architecture definition, arithmetic consistency	This baseline
L1	First-order simulation / screening	SIM-001
L2	Component / subsystem bench	Tower bench, ACT-001
L3	HIL / system integration	State-machine HIL, power-loss timing
L4	Vehicle-level prototype	Instrumented mule
L5	Production-intent / certification	Beyond this baseline

Gate	Focus	Key exit criteria
G0	Document release	Owner release; reviewer appointed

Gate	Focus	Key exit criteria
G1	Architecture freeze / packaging	3-D datum stack validating the re-baselined stack with residual ≥ +20 mm or re-baselined heights (REQ-M32); design vertical impact speed fixed (REQ-M33); ICD-001 draft; REG-001; cycle definition; Class S study (REQ-M30); V-L pack rule applied to supplier data (REQ-M36 / REQ-E03)
G2	Subsystem evidence	Tower bench; SAF-001; supplier data; ACT / STR / ENE closure; stop-curve efficiency measured (REQ-M33); horizontal path sized (REQ-M34); seam function, hybrid firewall and electric vent isolation demonstrated (REQ-M35, REQ-H07, REQ-E02)
G3	System HIL	Full state machine; power-loss; lateral-g
G4	Vehicle mule	Instrumented range and structural tests
G5	Production readiness	Beyond this baseline

17. Gates, work packages, risks and open items

17.1 Open items

#	Item	Closure
1	Packaging re-baselined stack (0 / +10 mm) conditional on 170 mm of unvalidated levers (P0)	G1 3-D stack
2	P90 module mass four-tower (427–565 kg); supplier correlation	G2
3	Pack specific energy confirmation (250 vs 180–200 Wh/kg)	Supplier data
4	Cycle definition for ≥800 km claim	SIM-FE-001 / ENE-FE-001
5	Height sensitivity: mass, CG and CdA effect of +250 / +130 mm	G1 SIM / CFD
6	Lock / S4 logic from COMFORT	ICD-FE-001 / LCK-001
7	Full state-machine S5–S8 transitions	ICD-FE-001
8	Regulatory 3,500 kg threshold (V-L design GVW 4,044 kg)	REG-FE-001 G1
9	IP filing receipt	Counsel
10	Independent technical reviewer	≤2026-12-05 or G1
11	Energy buffer sizing both configs	ENE-FE-001
12	Stop stroke and micro-stage clearance (50 mm placeholder)	STR-FE-001 / G1
13	Structural corner distribution (CG, transfer)	STR-FE-001
14	High-severity pre-G2 review	SAF-FE-001
15	PARAM-M21-ROLL-GRAD	G2
16	V-L margin +156 kg (275 kWh) valid at 250 Wh/kg pack specific energy; −150 kg at 200 Wh/kg	G1 Class S study / supplier
17	Protection package inclusion policy	G1 mass closure
18	Class L four-tower stiffness (first mode ≈1.4 Hz)	ACT / STR-FE-001
19	Lever feasibility: free play 10 mm, 60 mm shared tower depth	ICD-FE-001 / STR-FE-001
20	Seam envelope between the moving cell and fixed bodywork (water, dust, noise, pinch)	G1 / ICD-FE-001
21	Lower bodywork (hood, wings, trunk, rear body) mass itemisation; currently inside the chassis group (A-15)	G1 mass freeze
22	Design vertical impact speed (baseline 3.0 m/s) and stop-curve efficiency η ≥ 0.70 (REQ-M33)	STR-FE-001 / G1
23	Horizontal crash path: capture-frame load bound at 20 g (REQ-M34)	STR-FE-001 / SAF-FE-001
24	V-L pack sizing rule at G1 (Appendix F.4, REQ-E03)	Owner / G1
25	Seam vent isolation for pack runaway gas (REQ-E02)	ICD-FE-001 / THERM-FE-001

17.2 Selected risks

ID	Risk	L	I	Score	Mitigation
R-22	Re-baselined stack closes only if all 170 mm of levers are delivered	4	4	16	G1 3-D datum stack
R-E3	V-L margin +156 kg (275 kWh) lost at 200 Wh/kg pack specific energy (−150 kg)	3	4	12	Supplier data; G1 Class S study
R-E6	V-H four-tower 20 g corner load ≈101 kN (hybrid ≈99 kN)	2	4	8	STR-FE-001; payload policy
R-E4	V-H 800 km not met at highway speed	4	3	12	Cycle definition; accept utility
R-E5	Regulatory 3,500 kg	3	3	9	REG-FE-001
R-E7	Height increase raises CdA and mass; ranges move	3	3	9	G1 SIM / CFD
R-E8	Seams between the moving cell and fixed bodywork not designed (water, dust, noise, pinch)	3	3	9	ICD-FE-001 / G1
R-E9	Stop-curve efficiency below 0.70 or design impact speed above 3.7 m/s: the 50 mm stroke is not enough and the stack grows	3	4	12	STR-FE-001
R-E10	Horizontal crash path into the cockpit unsized (≈58 / 101 kN per link at 20 g)	3	4	12	STR-FE-001 / SAF-FE-001

18. Controlled document tree

Master MCRS-DOC-00-R8.09 governs all module values. This document (R7.09) is the controlling electric-host baseline and is subordinate to the Master for module parameters. Hybrid host is MCRS-DOC-00-R5.40.

Document / WP	Role	Gate
MCRS-DOC-00-R8.09	Master — module values	Owner
MCRS-FE-DOC-00-R7.09	Electric host baseline (this)	Owner
MCRS-DOC-00-R5.40	Hybrid host baseline	Owner
ICD-FE-001	Interface control	G1 draft
SIM-FE-001	System simulation	G1
ENE-FE-001	Energy / range / buffer	G1/G2
SAF-FE-001	Safety / HARA	G2
STR-FE-001	Structural / stops	G2
ACT-FE-001	Actuator characterisation	G2
REG-FE-001	Regulatory	G1
THERM-FE-001	Thermal	G2
LCK-001	Lock logic (shared)	G1/G2

19. IP, disclosure and commercial paths

Founder / Legal Owner: Amirbahador Atai. Licensable module: MCRS-MOD-001. Paths A / B / C as in the Master. Wording: integration into a two-piece vehicle body. Trade-mark: MCRS. No filing receipt; provisional under counsel OPEN. Distribution limited to parties under a valid NDA.

20. Claims discipline, glossary and release

Claim type	Allowed statement	Forbidden statement
Range (electric)	“≥800 km design capability under defined cycle (STUDY)”	“800 km verified / certified range”
Range (hybrid)	“1,000 km target for cycle ≤70 km/h avg (STUDY)”	“1,000 km verified range at any speed”
Energy	“275 kWh (V-L) / 300 kWh (V-H) usable TARGET”	“certified capacity”
Safety	“Invariants defined; screening FMEA at L0”	“ASIL-rated”, “crash-certified”
Comfort	“Targeted isolation (unproven at L0)”	“Superior ride”, clinical outcome
Mass	“P50 closed with reserve; P90 OPEN”	“Mass closed”, “production mass”
Packaging	“FIXED-FOR-ENGAGEMENT / OPEN-P0”	“Locked geometry”, “packaging closed”
Readiness	“TRL 2–3, evidence L0”	“Production ready”, “homologated”
Actuator	“Km ≥63 N/√W ASSUMPTION”	“Proven force capability”

Term	Definition
CMFSB	Captured Mechanical Floating Safety Body — the isolated upper safety cell architecture
MCRS / MCRS-MOD-001	Modular Captured Ride System (product family) / the licensable four- or six-tower module
COMFORT (S2) / SPORT (S2S) / S4	Isolation mode / locked mode with all required locks confirmed / power-loss capture
Class S / M / L	Module corner mass and load class (light / intermediate / heavy host)
P50 / P90 / RSS	Percentile mass estimates; root-sum-square combination of independent deviations
FIXED-FOR-ENGAGEMENT / OPEN-P0	Locked for packaging studies but not closed / blocking packaging item
L0–L5 / G0–G5	Evidence ladder levels / programme gates
BCE / HVIL	MCRS Body Control ECU / High-Voltage Interlock Loop
Usable energy	Net energy available after the reserved / inaccessible fraction (≈10 %)
RA-1	Ambulance / specialty medical transport (comfort context only; no clinical claim)

Appendix A — Correction log

A.1 R7.03 → R7.04 (retained)

#	Item	Action in R7.04
HA-01	Corrected and Fixed passes shared ID R7.03; Master reference R8.02	New ID R7.04; cites Master R8.04
HA-02	P90 wording “pump linear” did not reproduce 605 kg	§8 states exact method and sensitivity; V-H now four-tower 427 / 439 / 565 kg
HA-03	V-H utility band 880–1,000 km and V-L 120 km/h band optimistic	§7.3 recomputed (with new heights and masses)
HA-04	Class S distribution allocation not stated	§8 states 280 kg corner-only and the +96 to +113 kg exposure
HA-05	CFG-FE-001 undefined; R-01 ambiguous; exceeded ceiling not tagged	References removed; ceiling now 4,200 kg with +48 kg margin at that issue (+156 kg since D-06)
D-01	Owner decision: V-L ceiling 4,200 kg	V-L margin +48 kg at that issue (+156 kg since D-06); REG note updated
D-02	Owner decision: four V-H towers	Module 565 → 393 kg; curb 5,695 kg; payload residual 705 kg; upper mass 2,755 kg; 20 g corner ≈135 kN; mode ≈1.2 Hz; R-BATT retired
D-03	Owner decision: heights 1,950 / 2,130 mm	Stack 0 / +10 mm conditional; CdA scaled; ranges recomputed
CL-09	Same pack, different DC fast-charge class	Both ≥250 kW class

A.2 Energy and mass change log (R7.02 → R7.09)

Parameter	R7.02	R7.09	Delta
V-L usable energy	216 kWh	275 kWh	+59 kWh
V-L pack mass @ 250 Wh/kg	≈960 kg	≈1,224 kg	+≈264 kg
V-L curb target	3,200 kg	3,544 kg	+344 kg
V-H curb target	5,400 kg	5,695 kg	+295 kg
V-L design GVW	3,700 kg	4,044 kg	+344 kg
V-L ceiling / margin	4,000 / 300 kg	4,200 / 156 kg	ceiling +200; margin −144
V-L range at 100 km/h	≈900 km (216 kWh, 240 Wh/km)	≈1,060 km (259 Wh/km at new CdA)	Improved

A.3 R7.04 → R7.05 (retained)

#	Item	Action in R7.05
D-05	Owner decision 2026-10-12: only about half of the body (the passenger cell) is suspended; the 300 kWh battery and powertrain are in the lower chassis and add no weight on the upper level	§3 mirrors Master §3.4: zones, side-view figure, seam envelope, payload-location rule; REQ-M31; open item 20; risk R-E8
CL-11	Battery relationship unclear	Stated: the battery loads neither the cell nor the module; it counts only in vehicle weight vs the 4,200 kg ceiling and, by thickness, in the 280 / 320 mm chassis stack height
CL-12	Payload implicitly assumed in the cell	Conservative baseline kept; payload in the rear lower zone is a sensitivity (V-H upper mass ≈2,050 kg, 20 g corner ≈101 kN, mode ≈1.4 Hz)

A.4 R7.05 → R7.06 (retained)

#	Item	Action in R7.06
D-06	Owner decision 2026-10-12: V-L battery 275 kWh usable (≈306 kWh gross, pack 1,224 kg at 250 Wh/kg); V-H stays 300 kWh	V-L curb 3,544 kg, GVW 4,044 kg, margin +156 kg (+43 kg with Class S 393 kg; −150 kg at 200 Wh/kg); V-L ≥800 km to ≈126 km/h; AC 12.5 / 6.3 h; V-H unchanged (≥800 km to ≈72 km/h; at 275 kWh only to ≈63 km/h)
D-07	Owner decision: only the safety cockpit is suspended; hood and trunk are attached to the lower body; payload rides in the trunk	Upper sprung mass V-L ≈1,180 kg, V-H ≈2,050 kg (with protection); 20 g corner ≈58 / 101 kN; first mode ≈1.9 / 1.4 Hz; payload-location open item closed
CL-13	Lower bodywork mass not itemised	Assumption A-15: inside the chassis group; itemise at G1 mass freeze (open item 21)

A.5 R7.06 → R7.07 (retained)

#	Item	Where recorded
EC-03, EC-05, EC-07, EC-09, EC-10	Consistency defects found in R7.06	Appendix F.1
EN-01 to EN-07	Engineering findings of the R7.07 review	Appendix F.2

A.6 R7.07 → R7.08 (retained)

#	Item	Action in R7.08
RV-01 to RV-12	Review findings of R7.09	Appendix F.3; tables F.4 to F.8
Page rule	New page count at least the previous page count	R7.08 has 31 pages against 29 in R7.07

A.7 R7.08 → R7.09 (this issue)

#	Item	Action in R7.09
RX-01 to RX-06	Review findings of R7.09	Appendix F.9; Appendix G
Page rule	New page count at least the previous page count	R7.09 has 33 pages against 31 in R7.08

Appendix B — Status tag definitions (normative)

Tag	Meaning	Change control
BASELINE	Architecture decision frozen for this issue	Requires revision and owner release
TARGET	Design intent under evidence	G1/G2 evidence can refine
STUDY	Screening only; not design commitment	Superseded by supplier / test data
FIXED-FOR-ENGAGEMENT	Locked for packaging studies; not packaging-closed	Re-baseline at G1 if stack fails
DERIVED	Arithmetic consequence of controlled values	Recalculated when inputs change
OPEN — P0	Blocking packaging open item	Must close or re-baseline before G1 exit
ASSUMPTION	Explicit assumption with owner and closure route	Closed by named work package
LOCKED	Hard ceiling or constraint	Programme policy change required
LOCKED–EXCEEDED	A LOCKED value that a controlled design value breaches	Open decision at the named gate; the breach must be shown wherever the value appears

Previous use of LOCKED for open packaging heights was misleading and is replaced by FIXED-FOR-ENGAGEMENT throughout.

B.1 Post-issue correction record (C-series)

- C-01 (R7.03): mass budget re-summed (V-L 3,652 kg; V-H 5,867 kg at that issue; superseded by D-02, D-06); GVW definition and status tags.
- C-02 (R7.04): ranges recomputed: reference masses 3,852 / 5,895 kg (V-L 3,852 kg superseded by D-06: 3,744 kg), CdA 0.99 / 1.60 m².
- C-03 (R7.04): mass budget re-summed after D-02: curb V-H 5,695 kg; payload residual 705 kg.
- C-04 (R7.04): structural loads recomputed for four towers with payload in the cell (1 g ≈6.8 kN; 20 g ≈135 kN; V-H mode ≈1.2 Hz); superseded by D-07 (§13: 5.0 kN; 101 kN; 1.4 Hz).
- C-05 (R7.05): body zoning and load boundary added (§3 mirror of Master §3.4); payload-location sensitivity added (superseded by D-07); references updated to Master R8.05 and Hybrid R5.36.
- C-06 (R7.06): V-L usable battery 275 kWh (V-H stays 300 kWh); only the safety cockpit is suspended, payload rides in the trunk; masses, ranges, charging, loads and modes recomputed; assumption A-15 added; references updated to Master R8.06 and Hybrid R5.37.
- C-07 (R7.07): review of R7.06 (EC-03, EC-05, EC-07, EC-09, EC-10; EN-01 to EN-07); REQ-E02 and REQ-E03 added; references updated to Master R8.07 and Hybrid R5.38.
- C-08 (R7.08): review RV-01 to RV-12 applied (stop-stroke efficiency and design speed, horizontal capture bound, V-L pack sizing rule, neutral ceiling wording, height / lever / seam sensitivities); R7.08 has 31 pages against 29 in R7.07; references updated to Master R8.08 and Hybrid R5.39.
- C-09 (R7.09): review RX-01 to RX-06 applied: G1 / G2 exit criteria include REQ-E02, REQ-E03 and REQ-M32 to REQ-M36 (§16); Appendix G added (release readiness and contract use); R7.09 has 33 pages against 31 in R7.08; references updated to Master R8.09 and Hybrid R5.40.

Appendix C — Work package summaries (electric host)

WP	Scope	Gate
SIM-FE-001	Multi-body and energy simulation under COMFORT and SPORT, power-loss sequences, road profiles; mode timing, force/stroke envelopes, range sensitivity to mass, CdA and height	G1
ENE-FE-001	Measured pack data; buffer energy and recharge for four- and six-tower peak power; charging curves AC 22/44 kW and DC	G1/G2
SAF-FE-001	HARA, FTA for capture and power-loss, independence argument, RPN rescoring, pre-G2 review of S ≥ 9	G2
STR-FE-001	Stop force–displacement, real corner load distribution, stop stroke, single-tower fail-safe path	G2
ACT-FE-001	Force–gap–thermal characterisation, Km, bandwidth, first mode	G2
ICD-FE-001	Mechanical / electrical / hydraulic / software interfaces; lock protocol; power-loss timing; lateral-g path	G1 draft / G2 freeze
REG-FE-001	Market exposure for V-L design GVW 4,044 kg (above 3,500 kg threshold in some jurisdictions), HV safety, crash	G1
THERM-FE-001	Pack, inverter and fast-charge heat rejection; cabin HVAC interaction; vent routing	G2
LCK-001	Lock confirmation, incomplete-set inhibit, power-loss behaviour, manual rescue (shared with hybrid)	G1/G2

Indicative timeline: Ph0a document release; Ph0b (months 1–3) SIM-FE-001 start, ICD-FE-001 draft, reviewer appointment, provisional IP filing; G1 (months 3–5) stack validation, cycle definition, REG-FE-001, Class S study; Ph1 (months 5–10) supplier engagement and bench planning; G2 (months 10–14) tower bench, SAF-FE-001, supplier pack data. The paid Option is a bounded subset of Ph0b/G1 deliverables. Times are indicative.

Appendix D — Parameter ID cross-reference

PARAM-ID	Description	Value / status	Freeze
PARAM-KM	Lorentz motor constant	≥63 N/√W at 2–5 mm gap (ASSUMPTION)	ACT-FE-001 G2
PARAM-REF-MASS-MAX	Reference test mass basis	Curb + 200 kg	SIM-FE-001
PARAM-TOWER-MASS-LIMIT	Core tower P90 limit	67 kg conservative until G2	G2 supplier
PARAM-MODE-TIME	COMFORT↔SPORT transition	≤3 s TARGET	HIL G3
PARAM-M21-ROLL-GRAD	Lateral-g filter / gradient	TBD	G2
PARAM-ENERGY-USABLE	Primary usable energy	275 kWh V-L / 300 kWh V-H	Owner
PARAM-PACK-SE	Pack specific energy gross	250 Wh/kg ASSUMPTION; 180–200 sensitivity	Supplier G1/G2
PARAM-P90-METHOD	P90 combination method	√(n × 223 + 16²) kg deviation	G2 supplier
PARAM-HEIGHT	SPORT overall height	1,950 / 2,130 mm (D-03)	G1 stack

Appendix E — Release and distribution control

This document is T1 Confidential / NDA controlled. Distribution is limited to parties under a valid NDA. Public disclosure of module architecture, energy claims beyond the claims-discipline table, or host performance statements is prohibited until owner release and, where applicable, IP filing.

Revision R7.09 supersedes R7.07 for all electric-host content. Module values remain under Master MCRS-DOC-00-R8.09. Hybrid-host content is controlled by MCRS-DOC-00-R5.40. Any conflict on module parameters is resolved in favour of the Master. The Independent Technical Reviewer is required no later than the earlier of 2026-12-05 and G1 closure, and before any external safety-performance credit. Approval authority is TBD after independent review.

Appendix F — Design and engineering review (R7.09)

This appendix records the review history of this document. Review R7.07 found consistency defects in R7.06 and engineering findings; review R7.08 found further defects in R7.07; review R7.09 found the gap described in F.9 (RX-01). Calculations are first-order (L0); they set bounds and closure criteria and do not replace evidence from SIM-FE-001, STR-FE-001, ICD-FE-001 or supplier data. Module-level requirements REQ-M32 to REQ-M36 are defined in Master R8.09 Appendix M. Owner decisions D-01 to D-07 are unchanged.

F.1 Consistency defects closed in R7.07 (retained)

ID	Finding in R7.06	Resolution in R7.07
EC-03	Appendix E stated that R7.06 supersedes R7.03.	Supersession restated; Master and Hybrid references updated.
EC-05	The zone table listed the electric battery as 300 kWh and the mass table kept the note “slight increase with 300 kWh packaging”.	Zone table corrected to 275 / 300 kWh (§3.4); mass-table note corrected (§8) in R7.09.
EC-07	The C-series record carried superseded values without a label and was out of order.	Records in ID order; superseded values labelled.
EC-09	New content issued under the existing ID R7.06.	Issued under a new ID; history, tree and supersession updated.
EC-10	Page count must not fall below the original issue.	R7.07 had 29 pages against 25; restated as the page rule for R7.09.

F.2 Design and engineering findings (electric host) and status in R7.09

ID	Finding	Resolution / requirement	Status in R7.09
EN-01	The stack closes at 0 mm (V-L) and +10 mm (V-H) with every lever unvalidated.	REQ-M32: residual ≥ +20 mm at G1 or re-baseline (V-L ≈1,970 mm, V-H ≈2,140 mm).	OPEN; consequences in table F.7
EN-02	The 50 mm stop stroke has no link to a design impact speed.	REQ-M33.	Refined by RV-01 (table F.6)
EN-03	The horizontal crash path into the cockpit is not sized.	REQ-M34.	Bound in table F.6; OPEN
EN-04	Seams have a motion requirement only; pack runaway venting must not reach the cell.	REQ-M35 and REQ-E02.	OPEN — ICD-FE-001 / THERM-FE-001
EN-05	Average coil power was not stated.	(700 ÷ 63)² ≈ 123 W per tower, ≈0.49 kW for four towers, inside the 3 kW DC-DC.	Closed
EN-06	Class S exposure is +113 kg; the electric V-L reserve (140 kg) covers it.	REQ-M30.	OPEN until the G1 Class S study
EN-07	The V-L margin is +156 kg at 250 Wh/kg, +43 kg with Class S, −263 kg with Class S at 200 Wh/kg.	REQ-M36 and REQ-E03.	Rule in table F.4 (RV-03); OPEN owner / G1

F.3 Review R7.08: findings in R7.07 and resolutions

ID	Finding in R7.07	Resolution in R7.08
RV-01	The arrest table assumed an ideal constant-force stop.	Efficiency η = 0.70 and baseline design speed 3.0 m/s (table F.6).
RV-03	The V-L sensitivities (−150 kg, −263 kg) had no closure rule.	Pack sizing rule (table F.4): 275 kWh holds for specific energy ≥241 Wh/kg (Class S 393 kg) or ≥221 Wh/kg (Class S 280 kg).
RV-04	The option table said “Breaches the ceiling”.	Neutral wording (table F.5).
RV-06	The horizontal capture-frame load was not bounded.	F = m·a at 20 g and 30 g (table F.6).
RV-07 / RV-08	Height, CdA and lever effects on range were not quantified.	Table F.7.
RV-09	Seams had no overtravel requirement.	107.5 mm overtravel; no pack-gas vent path (REQ-E02).
RV-10	The lower-bodywork mass (A-15) had no bound.	Margin +156 kg absorbs up to 156 kg (43 kg with Class S 393 kg); V-H payload residual 705 kg.
RV-11	The page rule was “must not fall below”.	R7.08 has 31 pages against 29 in R7.07.

F.4 Pack sizing rule for V-L electric (REQ-E03)

Everything except the pack and the Class S module weighs 2,320 kg (3,544 − 1,224). With occupants 400 kg and payload 100 kg the allowed pack mass is 1,267 kg (Class S 393 kg) or 1,380 kg (Class S 280 kg). Usable energy ≤ allowed pack mass × gross specific energy × 0.9, never above 275 kWh.

Supplier specific energy	Max usable, Class S 393 kg	Adopted usable energy	Range at 100 km/h / speed holding 800 km	Max usable, Class S 280 kg	Adopted usable energy
250 Wh/kg	285 kWh	275 kWh	≈1,050 km / ≈125 km/h	310 kWh	275 kWh
225 Wh/kg	257 kWh	257 kWh	≈970 km / ≈118 km/h	279 kWh	275 kWh
200 Wh/kg	228 kWh	228 kWh	≈870 km / ≈107 km/h	248 kWh	248 kWh
180 Wh/kg	205 kWh	205 kWh	≈780 km / ≈97 km/h	224 kWh	224 kWh

F.5 V-L electric ceiling sizing (4,200 kg; payload 100 kg)

Case	GVW kg	Margin kg	Note
275 kWh, 250 Wh/kg, Class S 280 kg (baseline)	4,044	+156	Baseline
+ Class S module at 393 kg	4,157	+43	REQ-M30 exposure
+ pack at 200 Wh/kg, Class S 280 kg	4,350	−150	Above the ceiling; owner decision at G1
+ both	4,463	−263	Above the ceiling; owner decision at G1

F.6 Vertical arrest and horizontal bound (electric hosts)

Item	V-L electric	V-H electric
Upper sprung mass	1,180 kg	2,050 kg
Energy per corner at 3.0 m/s	1.33 kJ	2.31 kJ
Stroke at 3.0 m/s, η = 0.70	33 mm (50 mm placeholder, 17 mm spare)	33 mm
Speed limit of the 50 mm placeholder (η = 0.70)	3.7 m/s (0.70 m drop)	3.7 m/s
Horizontal force at 20 g / per link (four)	232 kN / 58 kN	402 kN / 101 kN
Horizontal force at 30 g	347 kN	603 kN

Stroke $s = v^2 / (2 \times 196\ m/s^2 \times \eta)$; $F = m \cdot a$. The 3.0 m/s design speed, η = 0.70 and the 20 g mean pulse are planning assumptions that STR-FE-001 and SAF-FE-001 shall confirm or replace.

F.7 Height, CdA and lever sensitivity (electric)

Effect of +10 mm height	V-L (100 km/h)	V-H (90 km/h)
CdA increase	≈0.51 %	≈0.47 %
Consumption change	+0.29 %	+0.21 %
Range change	−0.29 %	−0.21 %

Lever not delivered	mm lost	Height needed V-L (residual 0)	Height needed V-H (residual 0)
Shared tower depth (60 mm)	60	2,010 mm	2,180 mm
Travel + free play reduction	30	1,980 mm	2,150 mm
Cabin H-point to roof	40	1,990 mm	2,170 mm
Floor-to-H-point	30	1,980 mm	2,150 mm
Shell thickness	10	1,960 mm	2,130 mm
All levers fail	170	2,120 mm	2,290 mm

F.8 Electric-host requirement additions

ID	Statement	Status
REQ-E02	Seam vent isolation: battery runaway gas shall not reach the cell through the seams (mirrors REQ-M35).	OPEN — ICD-FE-001 / THERM-FE-001
REQ-E03	V-L energy sizing: usable energy follows table F.4; design GVW with Class S and the specific-energy sensitivity ≤ 4,200 kg, or the owner re-decides at G1 (REQ-M36).	OPEN — Owner / G1
REQ-M32 to M36	Packaging margin, vertical arrest, horizontal crash path, seam function, ceiling sizing (Master Appendix M).	OPEN — Master App. M

F.9 Review R7.09: finding in R7.08 and resolutions

ID	Finding in R7.08	Resolution in R7.09
RX-01	The G1 and G2 exit criteria (§16) did not include REQ-E02, REQ-E03 and REQ-M32 to REQ-M36.	§16 mirrors the extended Master criteria: G1 adds REQ-M30, M32, M33, M36 and REQ-E03 (pack rule applied to supplier data); G2 adds REQ-M33 (stop-curve efficiency measured), M34, M35 and REQ-E02 (vent isolation).
RX-02 to RX-05	Assumptions, contract use and release conditions were not collected.	Master Appendix N and Appendix G of this document.
RX-06	Page rule.	R7.09 has 33 pages against 31 in R7.08.

Appendix G — Release readiness and contract use (electric host)

The general statement is in Master R8.09 Appendix N: this document may be used for NDA diligence, Path A Option scoping, host engagement studies and IP discussion under counsel; it must not be used for any warranty of range, charging time, safety, crashworthiness, homologation, mass or packaging, or for production or price commitments.

Electric-host item	Status	Closure
Owner release and confirmation of the register (Master N.1)	Pending	Owner
V-L 275 kWh / V-H 300 kWh and pack specific energy 250 Wh/kg	Assumption; no supplier data	Supplier data, G1 / G2
V-L ceiling margin +156 kg (+43 kg with Class S 393 kg)	Holds at 250 Wh/kg; rule F.4 otherwise	Owner / G1
Range figures (CdA scaled with height; no regeneration)	Screening only	SIM-FE-001 / ENE-FE-001
Pack runaway gas isolation from the cell (REQ-E02)	Open	ICD-FE-001 / THERM-FE-001
Stack closure (0 / +10 mm, conditional)	Open — P0	G1 3-D datum stack
Independent Technical Reviewer and counsel review	Not appointed	Owner

Role	Name	Date	Signature
Owner release (Legal Owner)	Amirbahador Atai	To be completed	To be completed
Independent Technical Reviewer	To be appointed	To be completed	To be completed

— END OF MCRS-FE-DOC-00-R7.09 —